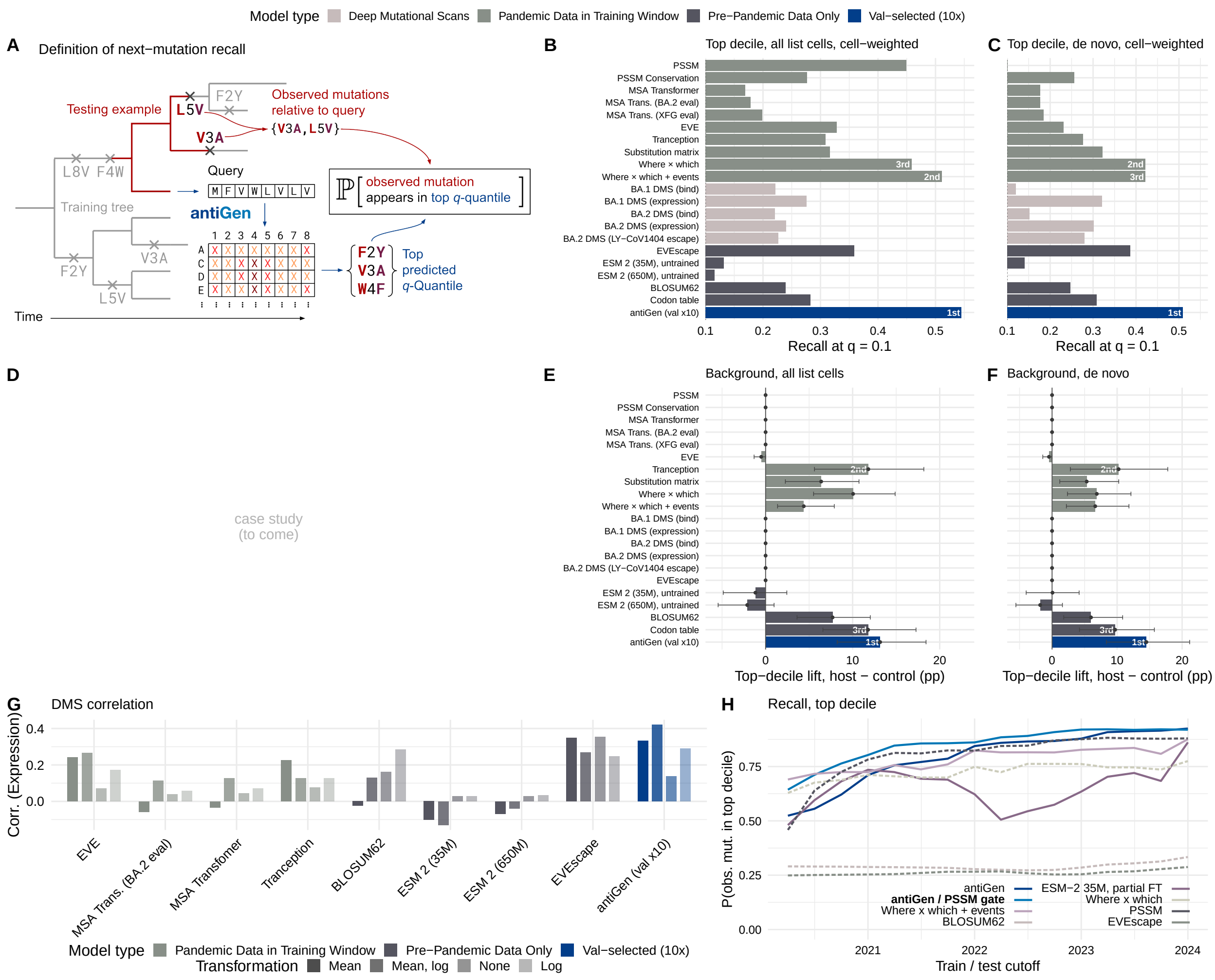

Model type
Deep Mutational Scans
Pandemic Data in Training Window
Pre−Pandemic Data Only
Val−selected (10x)
A
Definition of next−mutation recall
Testing example
L5V
F2Y
V3A
Observed mutations relative to query
{V3A, L5V}
L8V F4W
Query
M F V W L V L V
antiGen
ℙ[observed mutation appears in top q-quantile]
Training tree
V3A
F2Y
L5V
1 2 3 4 5 6 7 8
A
C
D
E
F2Y
V3A
W4F
Top predicted q-Quantile
Time
B
Top decile, all list cells, cell−weighted
PSSM
PSSM Conservation
MSA Transformer
MSA Trans. (BA.2 eval)
MSA Trans. (XFG eval)
EVE
Tranception
Substitution matrix
Where × which
Where × which + events
BA.1 DMS (bind)
BA.1 DMS (expression)
BA.2 DMS (bind)
BA.2 DMS (expression)
BA.2 DMS (LY−CoV1404 escape)
EVEscape
ESM 2 (35M), untrained
ESM 2 (650M), untrained
BLOSUM62
Codon table
antiGen (val x10)
3rd
2nd
1st
0.1 0.2 0.3 0.4 0.5
Recall at q = 0.1
C
Top decile, de novo, cell−weighted
2nd
3rd
1st
0.1 0.2 0.3 0.4 0.5
Recall at q = 0.1
D
case study
(to come)
E
Background, all list cells
2nd
3rd
1st
0 10 20
Top−decile lift, host − control (pp)
F
Background, de novo
2nd
3rd
1st
0 10 20
Top−decile lift, host − control (pp)
G
DMS correlation
Corr. (Expression)
0.4
0.2
0.0
EVE
MSA Trans. (BA.2 eval)
MSA Transfomer
Tranception
BLOSUM62
ESM 2 (35M)
ESM 2 (650M)
EVEscape
antiGen (val x10)
Model type
Pandemic Data in Training Window
Pre−Pandemic Data Only
Val−selected (10x)
Transformation
Mean
Mean, log
None
Log
H
Recall, top decile
P(obs. mut. in top decile)
0.75
0.50
0.25
0.00
2021
2022
2023
2024
Train / test cutoff
antiGen
antiGen / PSSM gate
Where x which + events
BLOSUM62
ESM−2 35M, partial FT
Where x which
PSSM
EVEscape